5.1 Angles of Triangles

Name _____ Period ____ Date _____

Geometry 5.1 (30-minute plan) · Student sheet

I can…

- classify triangles by their sides and by their angles.
- use the Triangle Sum Theorem to find angle measures.
- use the Exterior Angle Theorem to find angle measures.

Vocabulary · fill in the missing word

Scalene triangle — a triangle with no _____ sides

Isosceles triangle — a triangle with at least two _____ sides

Equilateral triangle — a triangle with three _____ sides

Exterior angle — an angle formed by one side of a polygon and the _____ of an adjacent side

Corollary — a _____ that can be proved easily using a theorem

Part 1 · Notes and examples (fill in each note, then work the example)

Classifying triangles: by sides: scalene, isosceles, equilateral. By angles: acute, right, obtuse, or _____ (all angles congruent).

Triangle Sum Theorem: the sum of the measures of the interior angles of a triangle is _____.

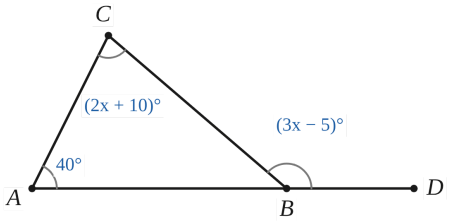
Example 1

Two angles of a triangle measure 52° and 71°. Find the third angle.

Exterior Angle Theorem: the measure of an exterior angle equals the _____ of the measures of the two nonadjacent interior angles.

Example 2

Find x and the measure of the exterior angle.



Corollary: the acute angles of a right triangle are _____.

Example 3

The acute angles of a right triangle measure $(2x)°$ and $(3x - 5)°$. Find both angles.

Part 2 · Partner practice

Partner practice

An exterior angle of a triangle measures 130°. One nonadjacent interior angle measures 72°. Find the other nonadjacent interior angle.